Exam on Array Algorithm

Marks: 60

(Solve any six)

Write Variable Description Table for at least one program

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1. Write a program to take N elements from user and sort them in ascending order using **bubble sort**. Print the sorted array and exact number of exchanges it required.

2. Using **linear search**, given an array of N integers and a key, find the index of **first and last occurrence** of the key (User input). If not found, print message accordingly

3. Write a program to take an array of size N, use **selection sort** to sort first K elements in ascending order and rest in descending order. (K is given by user)

4. Write a program to initialize the following data set in an array and find user given character's index using **binary search**. Print appropriate message if element is not found

   ['A', 'D', 'E', 'G', 'I', 'J', 'K', 'Y', 'Z']

5. Write a program to take N elements from user in an array and **remove duplicates**. Print the new array

6. Write a program to take N elements from user in an array and store **duplicate elements** in another array such that the new array should store one copy of each duplicate element.

7. Write a program to take 20 elements from user in an array. **Reverse** the array without using a second array.

8. Write a program to initialize the given array and **delete** m (given by user) from the array if it exists in the array. Print the updated array.

   [12, 43, 32, 54, 45, 23, 78, 23, 19, 70, 100]

9. Make a function int binarySearch(int a[], int first, int last, int s) to return index of s

   Array is sorted but rotated → [5,6,7,8,1,2,3] . Find the index of a given number.

   Hint: find pivot (smallest element). Based on the user given element's position, call the binary search function with required parameters.